

Workshop 5

1. A blood pressure drug trial examined the change in blood pressure, y_i , over 6 weeks of therapy for hypertensive patients aged 40–45. Patients were given one of 3 drug formulations or a placebo, each at one of three dose rates. The data for 120 patients were modelled as

$$\mathbb{E}(y_i) = \mu + \alpha_j + \beta_k, \text{ if patient } i \text{ received formulation } j \text{ and dose group } k,$$

with the y_i assumed to be independent gamma random variables. Model checking plots suggested that the full model assumptions were satisfied, and the model was refitted without the α_j terms. The deviance for the full model was 193, while the deviance for the reduced model was 201. What hypothesis can you test using this information? What is the associated p-value and what do you conclude?

2. If you add a completely irrelevant predictor variable (i.e., one with true coefficient zero) to a GLM, introducing one additional parameter, how much decrease would you expect in the scaled deviance?
3. This question concerns the Belgian AIDS epidemic data discussed in Section 10.1 of the lecture notes. The data can be entered into R as follows:

```
y <- c(12, 14, 33, 50, 67, 74, 123, 141, 165, 204, 253, 246, 240)
t <- 1:13
```

Recall that y denotes the number of new AIDS cases per year and t denotes the number of years since 1980 (up to and including 1993).

- (a) Plot the number of AIDS cases against year since 1980.
 - (b) Use the `glm` function to fit the model for the AIDS data given in the notes. You will need to use the `poisson` family with a `log` link function.
 - (c) Based on the scaled deviance, does the model appear to be adequate?
 - (d) Examine the residuals for the fitted model. Is the model adequate?
 - (e) Propose an alternative model for the data and fit it.
 - (f) Check the alternative model and determine whether all its terms are significant using the `summary` command.
 - (g) Use AIC to compare your revised model with the original model.
 - (h) Which model do you think best describes the data?
 - (i) What could you conclude about whether the epidemic was continuing to increase exponentially by the end of the data series?
4. This question concerns the O-rings data introduced in Question 4 of Workshop Sheet 4. The data are available on Learn in the file `orings.rda` and can be read into R as follows.

```
load("orings.rda")
```

Recall the context: in January 1986 the space shuttle Challenger exploded shortly after take-off. Subsequent investigation eventually focused on the possibility that O-rings in the fuel tanks had failed as a result of the unusually low launch temperature (31 degrees F), thereby allowing a fuel leak. You can read more in Wikipedia:

https://en.wikipedia.org/wiki/Space_Shuttle_Challenger_disaster

It turned out that data on O-ring failure in previous launches were available for various launch temperatures. In particular, for each temperature considered, we have the number of O-rings, out of 6, that failed or were damaged. As discussed in the solutions of workshop 4, a logistic regression model might be appropriate. Viewing each of the 6 O-rings at each launch temperature as independent binomial trials, and modelling the probability of failure as a function of temperature using a logit link function, leads to the logistic regression model

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \beta_1 + \beta_2 \text{temp}_i, \quad y_i \stackrel{\text{indep.}}{\sim} \text{Binomial}(6, p_i), \quad (1)$$

where p_i denotes the probability of O-ring failure/damage at temperature temp_i and y_i denotes the number of damaged o-rings, out of 6, at temperature temp_i .

- (a) Reproduce the plot from Workshop Sheet 4.
- (b) Fit the model suggested in (1) in R. You will need to use the `binomial` family and `logit` link function.
- (c) The standard residual plots for this model are of limited usefulness. Explain why.
- (d) To assess the plausibility of the model, overlay a curve showing the expected number of failures as a function of temperature on your original data plot. Also plot approximate 95% confidence limits for the mean number of failures at each temperature. Note that an inverse function of the logit link can be created in R as follows:

```
ilogit <- binomial()$linkinv
```

- (e) Obtain a point estimate and an approximate 95% CI for the expected number of O-ring failures at the Challenger launch temperature of 31 degrees F. What does this suggest? How reliable are the results likely to be?
 - (f) In some respects, the first observation in the `orings` dataset appears to be out of line with the others and is an extremely influential point in the fit. Use the `cooks.distance` function applied to your fitted model object to confirm this. What do you think should be done about this?
5. In this question, we will analyse the data in the *contingency table* from Question 3 in Workshop Sheet 4. Contingency tables arise when observations are classified according to two or more factor variables, and the interest lies in the number of observations falling into each combination of factor levels. The main question of interest is usually whether the factors are associated.

Question 3 on Workshop Sheet 4 considered a simple contingency table used to investigate whether there is an association between gender and belief in life after death. We can address this question by

using an analysis of deviance to compare the fit of two competing models of these data: one in which belief is modelled as independent of gender and another in which belief depends on gender (i.e., a model including a gender–belief interaction term). The data are as follows:

	Believer	Non-Believer
Female	435	147
Male	375	134

The data can be entered in R as follows (where 1 denotes a believer and 0 a non-believer).

```
al <- data.frame(y = c(435, 147, 375, 134),
                 gender = as.factor(c("F", "F", "M", "M")),
                 belief = as.factor(c(1, 0, 1, 0)))
```

As we have seen, under the null model of independence between belief and gender, if y_i is the observed count in one of the cells of the table, then the expected count can be written as

$$\mu_i \equiv \mathbb{E}(Y_i) = n\gamma_k\alpha_j \text{ if } y_i \text{ is data for gender } k, \text{ and faith } j.$$

where n is the total number of individuals sampled, γ_1 and γ_2 are the proportions of females and males in the population sampled, and α_1 and α_2 are the proportions of believers and non-believers in the population sampled, respectively. Taking logs gives the linear predictor

$$\log(\mu_i) = \log(n) + \log(\gamma_k) + \log(\alpha_j) \text{ if } y_i \text{ is gender } k \text{ and faith } j.$$

Defining $\tau = \log(n)$, $\beta_k = \log(\gamma_k)$ and $\delta_j = \log(\alpha_j)$, we get

$$\log(\mu_i) = \tau + \beta_k + \delta_j \text{ if } y_i \text{ is gender } k \text{ and faith } j.$$

Note that the appropriate distribution for the data is technically multinomial. However, the likelihood is equivalent to the likelihood obtained by assuming that the counts in each cell of the table are observations from Poisson random variables. Therefore, the Poisson distribution can be used in practice.

- (a) Fit the model assuming that belief and gender are independent. Since there are only four observations and only one residual degree of freedom, there is no point in looking at residual plots. Based on the scaled deviance, does the model appear to be adequate?
 - (b) Do these data provide any evidence for an association between gender and belief in an afterlife?
6. This exercise involves data from a dose-response study concerning the number of adult flour beetles that died after five hours of exposure to gaseous carbon disulfide at various dosage levels. The data can be entered into R as follows:

```
logdose <- c(1.691, 1.724, 1.755, 1.784, 1.811, 1.837, 1.861, 1.884)
dead <- c(6, 13, 18, 28, 52, 53, 61, 60)
n <- c(59, 60, 62, 56, 63, 59, 62, 60)
```

The goal of this exercise is to fit a binomial regression model using three different link functions and assess which one provides the best fit to the data. Let p_i be the probability of death at (log) dose level x_i . The three link functions to be considered are:

- *Logit link function:*

$$\log\left(\frac{p_i}{1-p_i}\right) = \eta_i = \beta_0 + \beta_1 x_i.$$

- *Probit link function:*

$$\Phi^{-1}(p_i) = \eta_i = \beta_0 + \beta_1 x_i,$$

where Φ denotes the standard normal cumulative distribution function. In R, please specify `family = binomial(link = "probit")`.

- *Complementary log-log link function:*

$$\log(-\log(1-p_i)) = \eta_i = \beta_0 + \beta_1 x_i.$$

In R, please specify `family = binomial(link = "cloglog")`.

7. The dataset `africa` from the `faraway` package in R contains information about the number of military coups in sub-Saharan Africa together with various political and geographical variables.¹. Namely, the following variables are available:

- `miltcoup`: number of successful military coups from independence to 1989.
- `oligarchy`: number of years the country was ruled by a military oligarchy from independence to 1989.
- `pollib`: Political liberalization - 0 = no civil rights for political expression, 1 = limited civil rights for expression but right to form political parties, 2 = full civil rights.
- `parties`: Number of legal political parties.
- `pctvote`: Percent voting in the last election.
- `popn`: Population in millions in 1989.
- `size`: Area in 1000 square km.
- `numelec`: Total number of legislative and presidential elections.
- `numregim`: Number of regime types.

The R session below attempts to build a model for the number of military coups based on the other variables.

- (a) Write short notes to accompany the R session, explaining what is being done statistically and which model you would finally choose for these data.
- (b) Interpret the coefficients of the final selected model. Do they make sense?
- (c) What is the proportion deviance explained for the selected model?
- (d) Is the model a useful predictive model for the number of coups?
- (e) From the fitted models, which model would you have selected if using AIC for model selection?

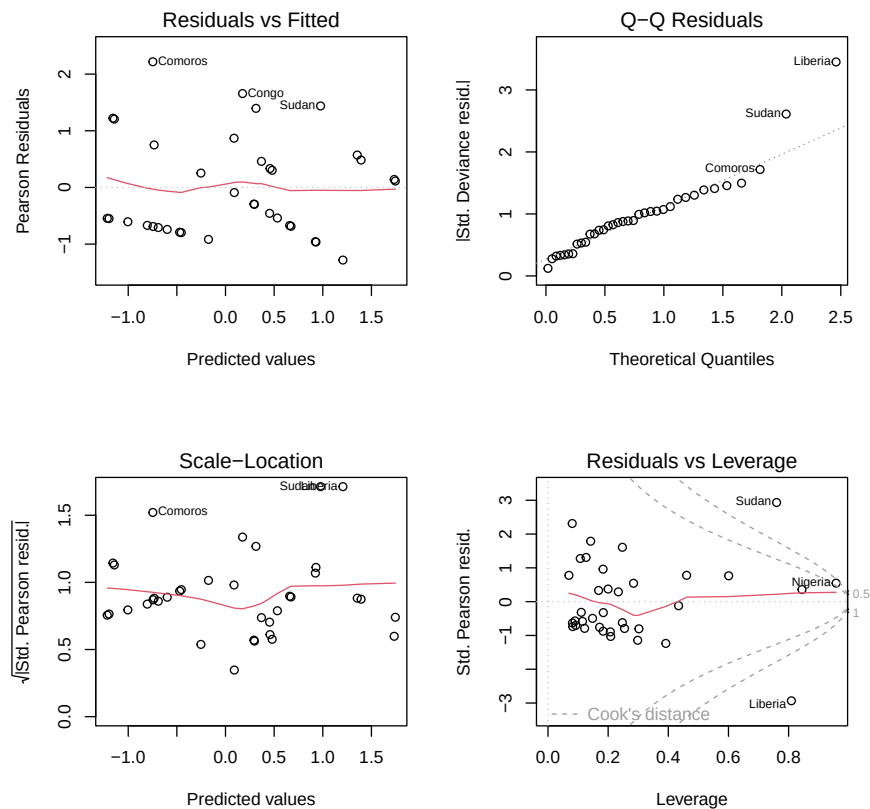
¹The post-independence history of this region is shaped by complex colonial legacies and external geopolitical pressures that cannot be captured by simple numerical predictors. The purpose of this analysis is strictly to illustrate the technical application of generalized linear models to count data and the process of model selection; the resulting models should therefore be interpreted as descriptive associations rather than causal explanations of political instability.

```

library(faraway)
data(africa)
ac1 <- glm(miltcoup ~ oligarchy + as.factor(pollib) + parties +
           pctvote + popn + size + numelec + numregim,
           family = poisson, data = africa)

par(mfrow = c(2,2))
plot(ac1)

```



```

drop1(ac1, test = "Chisq")

## Single term deletions
##
## Model:
## miltcoup ~ oligarchy + as.factor(pollib) + parties + pctvote +
##      popn + size + numelec + numregim
##           Df Deviance   AIC    LRT Pr(>Chi)
## <none>           28.249 113.06
## oligarchy         1   32.354 115.16 4.1045 0.042769 *
## as.factor(pollib)  2   35.581 116.39 7.3314 0.025587 *
## parties           1   35.311 118.12 7.0616 0.007875 **
## pctvote           1   30.572 113.38 2.3222 0.127537

```

```
## popn          1    30.601 113.41 2.3513 0.125178
## size          1    29.238 112.05 0.9881 0.320197
## numelec       1    28.430 111.24 0.1810 0.670499
## numregim      1    29.059 111.87 0.8099 0.368147
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ac2 <- update(ac1, .~. - numelec)
drop1(ac2, test = "Chisq")

## Single term deletions
##
## Model:
## miltcoup ~ oligarchy + as.factor(pollib) + parties + pctvote +
##      popn + size + numregim
##              Df Deviance    AIC    LRT Pr(>Chi)
## <none>                28.430 111.24
## oligarchy             1   36.595 117.40 8.1649 0.004271 **
## as.factor(pollib)     2   36.872 115.68 8.4417 0.014686 *
## parties               1   35.773 116.58 7.3428 0.006733 **
## pctvote              1   30.590 111.40 2.1600 0.141648
## popn                 1   30.605 111.41 2.1746 0.140305
## size                 1   29.452 110.26 1.0214 0.312196
## numregim             1   29.081 109.89 0.6508 0.419818
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ac3 <- update(ac2, .~. - numregim)
drop1(ac3, test = "Chisq")

## Single term deletions
##
## Model:
## miltcoup ~ oligarchy + as.factor(pollib) + parties + pctvote +
##      popn + size
##              Df Deviance    AIC    LRT Pr(>Chi)
## <none>                29.081 109.89
## oligarchy             1   40.291 119.10 11.2094 0.0008139 ***
## as.factor(pollib)     2   37.830 114.64  8.7487 0.0125960 *
## parties               1   36.304 115.11  7.2224 0.0071998 **
## pctvote              1   31.599 110.41  2.5175 0.1125903
## popn                 1   30.614 109.42  1.5324 0.2157552
## size                 1   30.040 108.85  0.9582 0.3276300
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ac4 <- update(ac3, .~. - size)
drop1(ac4, test = "Chisq")
```

```
## Single term deletions
##
## Model:
## miltcoup ~ oligarchy + as.factor(pollib) + parties + pctvote +
##      popn
##           Df Deviance    AIC      LRT Pr(>Chi)
## <none>           30.040 108.85
## oligarchy        1   40.468 117.28 10.4283 0.001241 **
## as.factor(pollib) 2   38.022 112.83  7.9826 0.018476 *
## parties          1   37.547 114.36  7.5070 0.006146 **
## pctvote          1   32.241 109.05  2.2010 0.137920
## popn             1   31.069 107.88  1.0299 0.310189
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ac5 <- update(ac4, .~. - popn)
drop1(ac5, test="Chisq")

## Single term deletions
##
## Model:
## miltcoup ~ oligarchy + as.factor(pollib) + parties + pctvote
##           Df Deviance    AIC      LRT Pr(>Chi)
## <none>           31.069 107.88
## oligarchy        1   48.196 123.00 17.1266 3.497e-05 ***
## as.factor(pollib) 2   39.762 112.57  8.6926  0.01295 *
## parties          1   37.547 112.36  6.4773  0.01093 *
## pctvote          1   32.822 107.63  1.7521  0.18561
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ac6 <- update(ac5, .~. - pctvote)
drop1(ac6, test="Chisq")

## Single term deletions
##
## Model:
## miltcoup ~ oligarchy + as.factor(pollib) + parties
##           Df Deviance    AIC      LRT Pr(>Chi)
## <none>           42.235 125.92
## oligarchy        1   64.574 146.25 22.3389 2.285e-06 ***
## as.factor(pollib) 2   48.597 128.28  6.3615  0.04155 *
## parties          1   45.431 127.11  3.1964  0.07380 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ac7 <- update(ac6, .~. - parties)
drop1(ac7, test = "Chisq")
```

```
## Single term deletions
##
## Model:
## miltcoup ~ oligarchy + as.factor(pollib)
##           Df Deviance   AIC      LRT  Pr(>Chi)
## <none>           45.431 127.11
## oligarchy         1   70.453 150.13 25.0219 5.668e-07 ***
## as.factor(pollib)  2   50.101 127.78  4.6694  0.09684 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(ac7)

##
## Call:
## glm(formula = miltcoup ~ oligarchy + as.factor(pollib), family = poisson,
##      data = africa)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      0.4795     0.4274   1.122   0.2618
## oligarchy         0.1023     0.0207   4.942 7.73e-07 ***
## as.factor(pollib)1 -0.5393     0.4695  -1.149   0.2506
## as.factor(pollib)2 -0.9168     0.4426  -2.071   0.0383 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 79.124  on 41  degrees of freedom
## Residual deviance: 45.431  on 38  degrees of freedom
## (5 observations deleted due to missingness)
## AIC: 127.11
##
## Number of Fisher Scoring iterations: 5
```

8. Suppose that Y follows a negative binomial distribution with probability mass function given by

$$f(y; \mu, k) = \frac{\Gamma(y+k)}{\Gamma(k)\Gamma(y+1)} \left(\frac{\mu}{\mu+k}\right)^y \left(\frac{k}{\mu+k}\right)^k, \quad y = 0, 1, 2, \dots$$

Further assume that $k > 0$ is a given constant and consider the random variable $Y^* = Y/k$. Then $\Pr(Y^* = y^*) = \Pr(Y = ky^*)$ and hence Y^* has probability mass function given by

$$f(y^*; \mu, k) = \frac{\Gamma(ky^* + k)}{\Gamma(k)\Gamma(ky^* + 1)} \left(\frac{\mu}{\mu+k}\right)^{ky^*} \left(\frac{k}{\mu+k}\right)^k, \quad y^* = 0, \frac{1}{k}, \frac{2}{k}, \dots \quad (2)$$

(a) Show that (2) belongs to the exponential family with

$$\theta = \log\left(\frac{\mu}{\mu+k}\right), \quad b(\theta) = -\log(1 - e^\theta), \quad a(\phi) = \frac{1}{k}.$$

- (b) Find the mean and variance of Y^* using expressions for $b(\theta)$ and $a(\phi)$. Use these results to show that $\mathbb{E}(Y) = \mu$ and $\text{var}(Y) = \mu + \frac{\mu^2}{k}$.
- (c) Compare the relationship between the mean and variance in the negative binomial distribution with that in the Poisson distribution, and comment on when the Poisson model is appropriate and when the negative binomial model is more suitable.